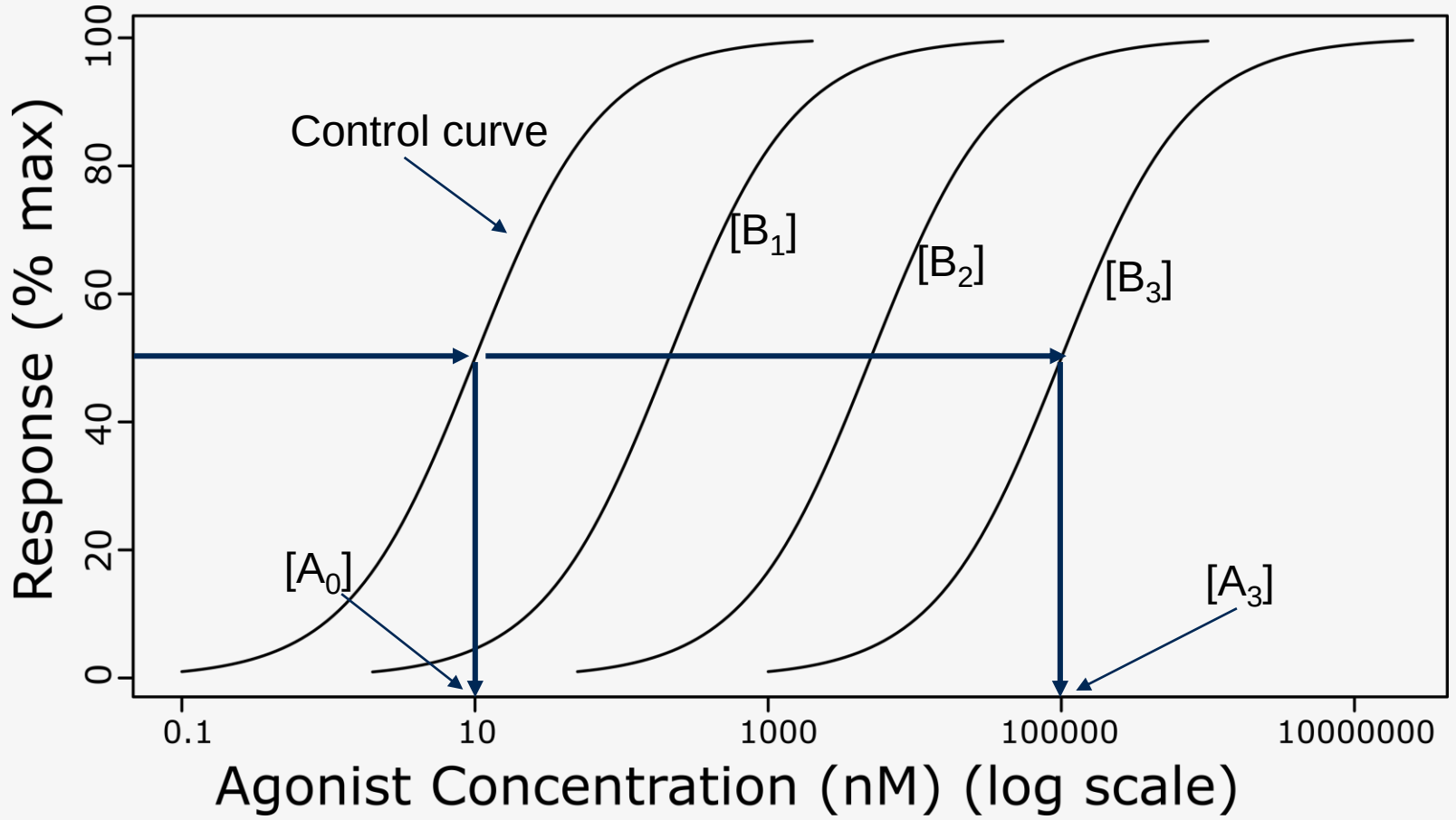
Looking for reversible competitive antagonism (continued).

$$p_{AR} = \frac{[\mathrm{A}]}{K_A\left(1 + \frac{[\mathrm{B}]}{K_B}\right) + [A]}$$

Gaddum Equation

Now we need to relate the Gaddum Equation for p_{AR} to our dose-response curves..



This helps because…

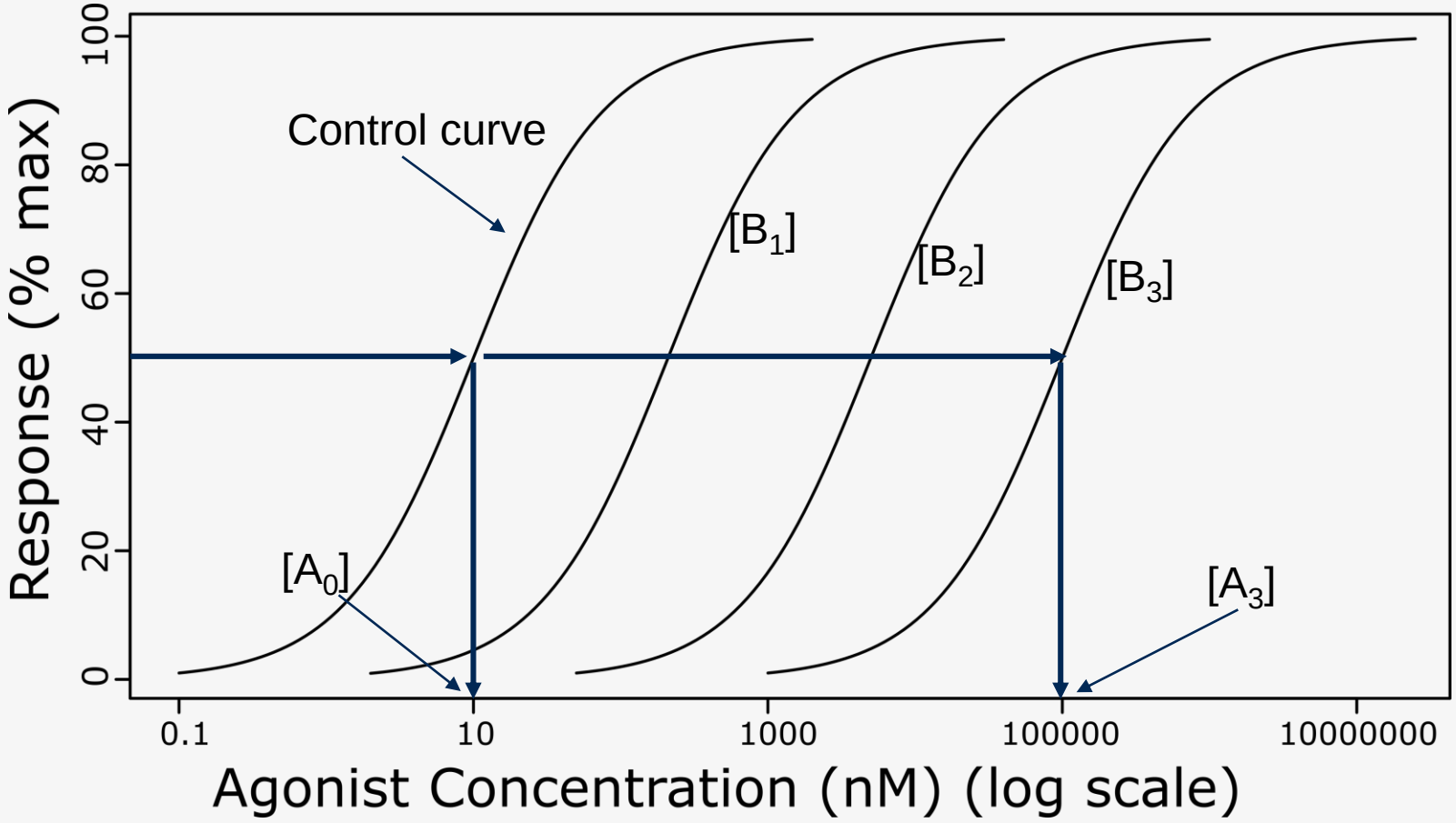
In the absence of antagonist, when [*B*]=0, and at 50% of the maximum response equation 15 becomes…

$$p_{AR} = \frac{[A_0]}{(K_A + [A_0])} \qquad \text{Eq. 17}$$

In the presence of antagonist, when [*B*]=[B_3], at the same 50% of the maximum response equation 15 becomes…

$$p_{AR} = \frac{[A_3]}{\left(K_A\left(1 + \frac{[B_3]}{K_B}\right) + [A_3]\right)} = \frac{[A_0]}{(K_A + [A_0])} \qquad \text{Eq. 18}$$

Now we need to relate the Gaddum Equation for p_{AR} to our dose-response curves..



Dose ratio 'r' such that e.g.

$$r_3 = \frac{[A_3]}{[A_0]}$$

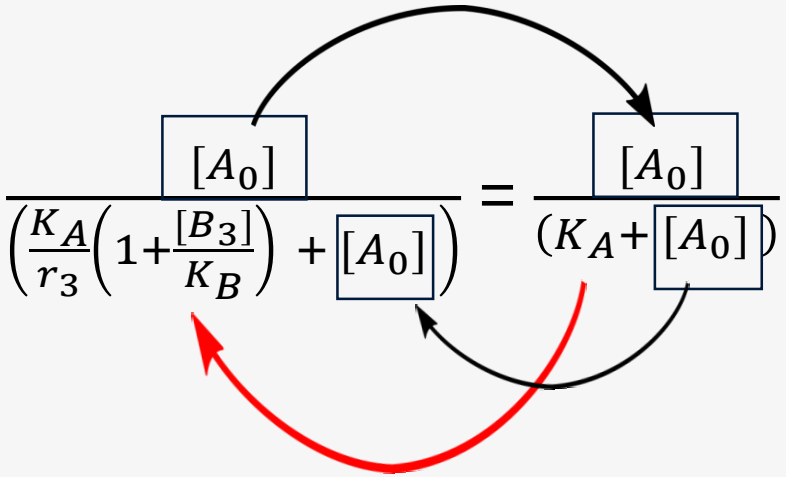
So… $r_3[A_0]=[A_3]$ Eq. 16

Substitute from equation 16 into equation 18 to get…

$$\frac{r_3[A_0]}{\left(K_A\left(1+\frac{[B_3]}{K_B}\right)+r_3[A_0]\right)} = \frac{[A_0]}{(K_A+[A_0])}$$

Eq. 19

Divide the top and bottom on the left-hand side by r_3



So… $K_A = \frac{K_A}{r_3}\left(1+\frac{[B_3]}{K_B}\right)$

And thus… $r_3 = 1+\frac{[B_3]}{K_B}$

The Schild Equation…

It always applies provided that:

1) The antagonist, B, is a true competitive antagonist (i.e not an inverse agonist or weak partial agonist).

2) At every binding site, the binding of agonist, A, and antagonist, B, is mutually exclusive.

3) The value of K_B is the same at every binding site.

4) When there is the same level of occupancy by A, there is always the same tissue response.

5) The system is at equilibrium.

And thus…

$$r = 1 + \frac{[B]}{K_B}$$

Is the Schild equation obeyed?

$$r = 1 + \frac{[B]}{K_B}$$

$$r - 1 = \frac{[B]}{K_B}$$

$$\underbrace{\log(r-1)}_{Y} = \underbrace{\log[B]}_{X} - logK_B$$

